

Enhanced Performance Of POWER ELECTRONICS With FASTER TESTING DEVICES

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This article describes the different types of measurement equipment required for power electronic systems

Performance of an electromechanical system is determined by the performance of individual power electronic components and switching devices. Hence, there is a need for accurate measurement of responses of power electronics, and other electrical and physical parameters. This article describes the different types of measurement equipment required for power electronic systems.

To select a measuring instrument for testing these systems, it is necessary to understand the testing parameters at various development stages. Test and measurement (T&M) equipment enable predictive and preventive inspections to minimise the risk of defects, accidents or electronic systems breakdowns. You can also derive optimum efficiency and operation for all kinds of troubleshooting.

Faster switching power electronic devices

The trend in power electronics is to push the operating frequencies higher to reduce size, weight and cost of the systems. This requires faster switching power electronic devices, such as SiC and GaN MOSFETS, and diodes. To test and characterise such devices, you need faster measurement devices like higher bandwidth oscilloscopes and power analysers.

For design and architecture stages, key factors that help select test instruments include high-frequency dynamic behaviour, static behaviour, fast inverter switching, trigger for individual waveforms and overshoot on pulses.

Measurement at this stage requires mixed-signal oscilloscopes with multiple channels.

Measured parameters

Dynamic behaviour of power electronics can be measured with a scopecorder (oscilloscope-cum-data acquisition recorder). A data acquisition recorder handles a wide range of power measurements and captures high-resolution details with a 12- or 16-bit ADC. A scopecorder carries out complex calculations to measure power factor, active power and harmonics using a digital signal processor.

Dynamic parameters such as switching times (T_{rise} , T_{fall} , T_{delay} or T_{rr}) use high-bandwidth oscilloscopes. Static parameter measurements need curve tracers and parametric analysers. Static parameters include V_{CEon} , V_{DS} , R_{DSon} , V_F , V_{GE} and leakage currents using curve tracers. Switching energies such as E_{on} , E_{off} and E_{rr} are measured using high-bandwidth current probes and oscilloscopes with math functions.

Quality control measures

Power analysis, conversion efficiency and harmonics are measured for efficiency validation. A power analyser measures efficiency, total harmonic distortion and power factor correction of circuits. For these, the test power analyser should be of high accuracy and high stability. It should have calibration ability that can measure highly distorted current and voltage waveforms accurately.

Online inspection and troubleshooting are important to reduce costs and downtime, and improve the efficiency of various electronic components. For online inspection, thermal imagers are useful for PCBs and SMD cards, for fault diagnosis and research purposes. This is due to their high IR resolutions of 320x240 pixels and 640x480 pixels. These can also be used for the mass production of electronic compo-



Fig. 1: Power electronic components: SiC-SBD (left) and SiC-DMOS (right) (Credit: www.rohm.com)



Fig. 2: Thermal imager (Credit: www.testo.com)